

Are LED TVs Better than LCD and Plasma TVs?

In terms of picture quality, LED TVs are better than LCD TVs, but these still don't match plasma TVs. For the same price, you can get a bigger-size LCD or plasma TV



■ UMA BANSAL

Planning to replace your good-old CRT TV with a brand new flat-panel TV? Or buying a flat-panel TV as part of your home theatre? Whatever be the case, when you are in a store to buy one, you are likely to get confused between LCD, plasma and the recently launched LED TVs. The three may look identical at a

glance but similarities disappear once you look past the thin form factor and big screens.

Which is better, LCD or plasma? Or, is it a safe bet to go with the new and expensive LED TV? To find out, let's compare some of their important technical features.

LCD vs plasma TVs

Contrast ratio. The dynamic contrast ratio helps the TV panel to reproduce

the natural images, by producing maximum possible colour shades. The higher the dynamic contrast ratio, the better the detailing of the image on the LCD TV.

Plasma TVs have higher contrast ratios than LCD panels. For example, LG plasma TVs are engineered to drive dynamic contrast ratio as high as 2,000,000:1 as against 150,000:1 for LCD TVs. The higher contrast ratio of plasma TVs results from their ability to turn the light source off for individual segments of the screen. This means parts of the screen can be completely dark while others display bright colour. By contrast, traditional LCD screens have a single backlight array that can only alter brightness levels for the entire screen.

Viewing angle. A wide viewing angle allows a big family to view the screen comfortably at the same time. The narrower the viewing angle, the smaller the range from which the TV screen can be viewed at acceptable image quality.

Typically, plasma TVs offer wider viewing angles than LCD TVs, with their image quality remaining unaffected at wide viewing angles while LCDs exhibit colour shift and loss of brightness.

Hitachi claims its LCD televisions offer an industry best viewing angle of 178 degrees, while its plasma televisions have no viewing angle problem (they offer free-viewing).

Response time. Response time is the pixel refresh rate (from black to